23A01601 · DESIGN OF STEEL STRUCTURES

UNIT – III

Design of Beams

Syllabus: Design of Laterally Supported and Unsupported Beams – Design of Built-Up Beams – Design of Plate Girders.

What's inside these notes

- Beam behaviour, section classification, bending & shear capacity
- Laterally supported beam – design steps & example
- Deflection, web buckling, web crippling (bearing)
- Lateral–torsional buckling & laterally unsupported beams (example)
- Built-up beams, cover plates, curtailment
- Plate girders: proportioning, flange & web design, stiffeners
- Plate girder worked example

Exam tip: Beam design order: (1) M_u, V_u → (2) trial section → (3) classify → (4) M_d (check LTB) → (5) shear → (6) deflection → (7) web buckling / crippling at supports.

Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take $E = 2\times10^5$ MPa, $f_y = 250$ MPa, $f_u = 410$ MPa unless stated.

1. Beams – Introduction

Beam = member carrying **transverse loads** (bending + shear). Types: simply supported, cantilever, continuous, fixed ; joists, girders (main), purlins, lintels, crane girders, floor beams.

Common sections: ISMB, ISLB, ISWB, ISMC, built-up I (plate girders), box, compound (I + plate / channel).

Failure modes of beams

- **Flexural yielding** (plastic hinge) – most economical
- **Lateral–torsional buckling (LTB)** – slender compression flange
- **Shear yielding / buckling of web**
- **Web crippling / buckling** under concentrated loads or reactions
- Local buckling of flange (slender section)
- Excess **deflection** (serviceability)

Section classification & ε (recap)

$\varepsilon = \sqrt{(250/f_y)}$. For rolled I-section (strong axis bending):

Class	Flange b/t_f (outstand)	Web d/t_w
Plastic	$\leq 9.4\ \varepsilon$	$\leq 84\ \varepsilon$
Compact	$\leq 10.5\ \varepsilon$	$\leq 105\ \varepsilon$
Semi-compact	$\leq 15.7\ \varepsilon$	$\leq 126\ \varepsilon$

Welded sections: flange 8.4ε / 9.4ε / 13.6ε.

2. Laterally Supported Beam – Bending Strength

Laterally supported = compression flange is **restrained** (slab, purlin sheeting with sufficient fixity, bracing) → no LTB; section yields.

Design bending strength (IS 800 Cl. 8.2.1.2)

$$M_d = \beta_b Z_p f_y / \gamma_{m0} \leq 1.2\, Z_e f_y / \gamma_{m0} \text{ (simply supported)}$$
$$\leq 1.5\, Z_e f_y / \gamma_{m0} \text{ (cantilever)}$$

- $\beta_b = 1.0$ for plastic & compact ; Z_e/Z_p **for semi-compact**
- Z_p = plastic section modulus ; Z_e = elastic section modulus
- Shape factor for I ≈ **1.12 – 1.15**, so limit $1.2\, Z_e f_y/\gamma_{m0}$ seldom governs.
- Slender (class 4): use effective section.

Shear strength (Cl. 8.4)

$$V_d = V_n/\gamma_{m0} ; V_n = A_v f_y/\sqrt{3}$$

$A_v = h\, t_w$ (rolled I, major axis, h = overall depth); $d\, t_w$ (welded, d = web depth).

- **Low shear** : $V \leq 0.6\, V_d$ → no reduction in M_d

- **High shear** : $V > 0.6\ V_d$ → reduced moment capacity:

$$M_{dv} = M_d - \beta\ (M_d - M_{fd}) \le 1.2\ Z_e\ f_y/\gamma_{m0}\ ;\ \beta = (2V/V_d - 1)^2$$

M_{fd} = plastic design strength of flanges alone.

> In most rolled sections with simple spans, shear is low except near short heavy loaded spans.

Design procedure (laterally supported beam)

- 1. Find factored M_u, V_u (1.5 × (DL+LL) ; include self-weight guess ≈ span/...).
- 2. Required $Z_p = M_u\ \gamma_{m0}/f_y$ (or $Z_e = Z_p$ / 1.14) → select ISMB/ISLB from tables.
- 3. **Classify** section (b/t_f, d/t_w); get M_d, check ≥ M_u.
- 4. Check **shear** $V_d \ge V_u$; low/high shear.
- 5. Check **deflection** (service load) $\delta \le$ limit.
- 6. Check **web buckling & crippling** at supports / concentrated loads.
- 7. Detail: bearing length, stiffeners if needed.

EXAMPLE 1 – SIMPLY SUPPORTED BEAM, L = 5 M, w_u = 40 KN/M (FACTORED), LATERALLY SUPPORTED

$M_u = 40 \times 5^2 / 8 =$ **125 kN·m** ; $V_u = 40 \times 5/2 =$ **100 kN**

$Z_p(req) = 125 \times 10^6 \times 1.1 / 250 = 550\ 000\ mm^3 = 550\ cm^3 \rightarrow Z_e \approx 550/1.14 = 482\ cm^3$

Try **ISMB 300** ($Z_e = 573.6\ cm^3$, $Z_p \approx 654\ cm^3$; $b_f = 140$, $t_f = 13.1$, $t_w = 7.7$, $R_1 = 14$, $I = 8603.6\ cm^4$)

Class: $b/t_f = 70/13.1 = 5.3 < 9.4$; $d = 300 - 2(13.1 + 14) = 245.8 \rightarrow d/t_w = 31.9 < 84$ → **Plastic**

$M_d = 1.0 \times 654 \times 10^3 \times 250/1.1 =$ **148.6 kN·m** ($\le 1.2\ Z_e\ f_y/\gamma_{m0} = 156.4$ ✓) > 125 ✓

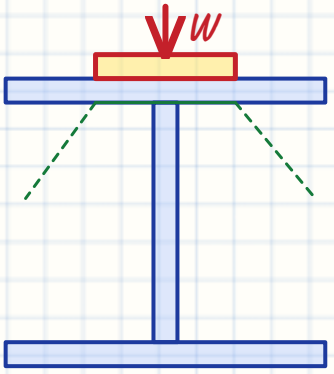
Shear: $V_d = 300 \times 7.7 \times 250/(\sqrt{3} \times 1.1) =$ **303 kN** > 100 ; $0.6\ V_d = 182 > 100$ → **low shear** ✓

Deflection (service w = 40/1.5 = 26.7 kN/m):

$\delta = 5wL^4/(384EI) = 5 \times 26.7 \times 5000^4/(384 \times 2 \times 10^5 \times 8603.6 \times 10^4) =$ **12.6 mm** < $L/300 = 16.7$ mm ✓

Use ISMB 300.

3. Web Buckling and Web Crippling (Concentrated Loads / Reactions)



1 : 2.5 dispersion through flange

→ stiff bearing length $b_1 + n_2$

web crippling $F_w = (b_1+n_2)\, t_w\, f_y/\gamma_{m0}$

web buckling $F_{cw} = (b_1+n_1)\, t_w\, f_{cd}$

(45° dispersion, $n_1 \approx d/2$ → column of web)

Dispersion of load through flange and web

Web crippling (bearing) – local yielding of web at toe

$$F_w = (b_1 + n_2)\, t_w\, f_{yw} / \gamma_{m0}$$

b_1 = stiff bearing length ; n_2 = length from dispersion at **1 : 2.5** through flange + root (to the web-flange junction).

Web buckling – web acts as a strut

$$F_{cw} = (b_1 + n_1)\, t_w\, f_{cd}$$

n_1 = dispersion at **45°** through half depth ; f_{cd} from column curve with slenderness $\lambda = 2.5\, d / t_w$ (for fixed flanges) – (effective length 0.7 d).

- If F_w or F_{cw} < reaction → provide **bearing stiffeners** or increase bearing length.
- Support reaction $R_u \le \min(F_w, F_{cw})$.

Web shear buckling

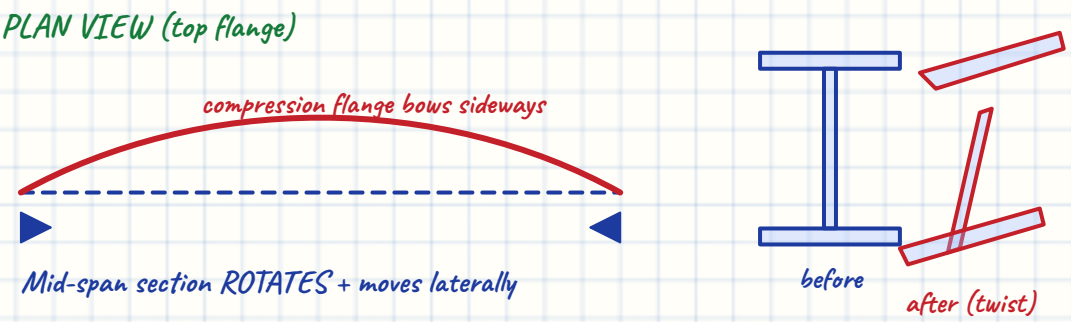
For $d/t_w \le 67\,\varepsilon$ → no check for shear buckling (rolled beams ok). For larger d/t_w : use **plate girder** shear provisions.

Deflection of beams (service load)

Case	δ_{max}
SS, UDL w	$5wL^4/384EI$
SS, central point load W	$WL^3/48EI$
Cantilever, end load W	$WL^3/3EI$
Cantilever, UDL	$wL^4/8EI$
Fixed, UDL	$wL^4/384EI$

Limits: L/300 floor beams, L/360 supporting plaster, L/150–L/180 purlins, L/750 crane girders.

4. Laterally Unsupported Beams – LTB



LTB is prevented by lateral restraint of the compression flange

Lateral–torsional buckling

When the compression flange is free to deflect sideways, the beam **buckles laterally + twists** (like a thin ruler bent flat) at a moment **less than M_p**. Strong-axis bending for slender (high I_{zz}/I_{yy}) sections.

Factors affecting LTB

- **Unbraced length L_{LT}** (↑ → ↓ strength) ; I_{yy}, torsional const. I_t, warping const. I_w
- Type of loading, **position of load** (top flange destabilising), support restraints
- Moment gradient, cross-section shape (box & tubes hardly LTB), residual stress

> LTB is **not a problem** for: box/hollow sections, bending about minor axis, beams fully restrained by slab, or if $\lambda_{LT} \le 0.4$.

Design bending strength (IS 800 Cl. 8.2.2)

$$M_d = \beta_b Z_p f_{bd} \; ; \; f_{bd} = \chi_{LT} f_y / \gamma_{m0}$$

$$\chi_{LT} = 1/[\phi_{LT} + \sqrt{(\phi_{LT}^2 - \lambda_{LT}^2)}] \le 1.0$$

$$\phi_{LT} = 0.5\,[1 + \alpha_{LT}(\lambda_{LT} - 0.2) + \lambda_{LT}^2]$$

$$\lambda_{LT} = \sqrt{(f_y / f_{cr,b})} = \sqrt{(\beta_b Z_p f_y / M_{cr})}$$

α_{LT} = **0.21** rolled sections, **0.49** welded sections.

Elastic critical moment (symmetric I-section)

$$M_{cr} = \sqrt{[(\pi^2 E I_y / L_{LT}^2)\,(G I_t + \pi^2 E I_w / L_{LT}^2)]}$$

Simplified (rolled/welded, symmetric I):

$$f_{cr,b} = [1.1\pi^2 E / (L_{LT}/r_y)^2]\sqrt{[1 + (1/20)\,((L_{LT}/r_y)/(h_f/t_f))^2]}$$

h_f = centre-to-centre of flanges ; $M_{cr} = Z_e f_{cr,b}$ … ; L_{LT} = effective unsupported length (end restraint conditions of Table 15).

Example 2 – ISMB 300 with L_{LT} = 4 m (unrestrained, simply supported)

r_y = 28.4 mm ; h_f = 300 − 13.1 = 286.9 ; h_f/t_f = 21.9 ; L_{LT}/r_y = 140.8

First term: $1.1\,\pi^2 E/(140.8)^2 = 1.1 \times 99.6 = 109.5$ MPa

Bracket : $1 + (1/20)(140.8/21.9)^2 = 1 + 2.067 = 3.067 \rightarrow \sqrt{} = 1.751$

$$f_{cr,b} = 109.5 \times 1.751 = 191.8 \text{ MPa}$$

$\lambda_{LT} = \sqrt{(250/191.8)} = \mathbf{1.142}$ (rolled $\alpha_{LT} = 0.21$)

$\varphi_{LT} = 0.5[1 + 0.21\,(1.142 - 0.2) + 1.142^2] = 0.5[1 + 0.198 + 1.304] = \mathbf{1.251}$

$\chi_{LT} = 1/[1.251 + \sqrt{(1.251^2 - 1.304)}] = 1/(1.251 + 0.511) = \mathbf{0.568}$

$f_{bd} = 0.568 \times 250/1.1 = \mathbf{129}$ MPa

$$M_d = 1.0 \times 654\times10^3 \times 129 = 84.4 \text{ kN·m (vs 148.6 kN·m when laterally supported!)}$$

So a 4 m unsupported length cuts capacity by ≈ 43 %.

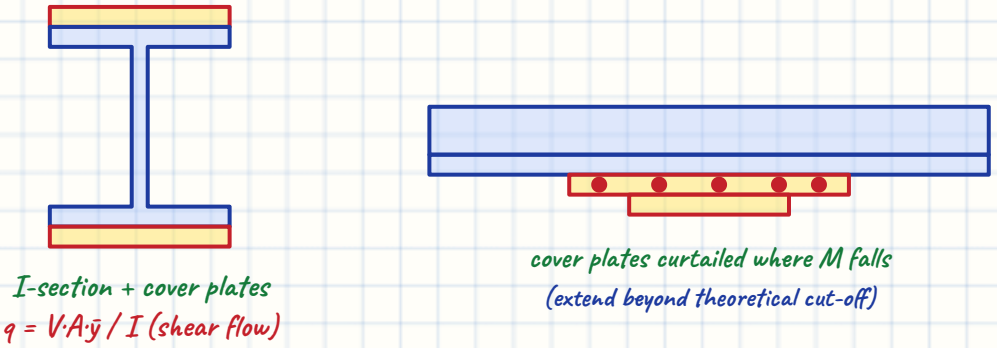
Ways to improve LTB strength

- Provide **lateral bracing / closer restraints** (cross beams, slab, diaphragms)
- Use sections with larger I_{yy} (ISWB, ISHB, box) ; add flange plate / channel on compression flange
- Lower the load point (bottom flange loading) ; ends restrained against twist

5. Built-up Beams (Compound Beams)

When a rolled section is **not enough** (heavy loads, long spans) → **strengthen** with

- **cover plates** welded/bolted to flanges ; extra **channel** on top flange ; **two I-sections** side by side ; **box** from plates ; **plate girder** (plates welded to form I).



Rolled beam with cover plates & curtailment

Design steps (I-section + cover plates)

- 1. Required Z_p ; rolled section provides part ; **extra Z from plates**: choose width b_p & thickness t_p ($b_p \le$ flange width, $t_p \le$ ~1.5 t_f total, outstand ≤ 16 t_p ε or 12 t) .
- 2. Compute I, Z_e of built-up section ; classify (cover plate outstand).

- 3. M_d, V_d, deflection, LTB as for rolled.
- 4. **Connection of cover plate to flange** – shear flow per unit length:

$$q = V A_p \bar{y} / I \ (A_p = \text{cover plate area}, \bar{y} = \text{centroid distance from N.A.})$$

- Pitch of bolts: $p = R_b / q$; or weld size from $q \le 2 \times f_{wd} t_t$ (two welds).
- 5. **Curtailment**: plate theoretical cut-off where $M_u \le M_d$ of rolled section alone. Extend by **length development** beyond cut-off: ≥ (flange width + development) so connection carries plate force. Plate end connected for its full force.
- Welds/bolts max pitch: 32 t or 300 (tension flange), 16t or 200 (compression).

EXAMPLE 3 – BOLT PITCH FOR COVER PLATE

$V = 300$ kN, $I = 25\,000$ cm^4 = 2.5×10^8 mm^4, cover plate 200 × 12 → $A = 2400$ mm², $\bar{y} = 250$ mm.

$q = 300\times10^3 \times 2400 \times 250 / 2.5\times10^8 =$ **720 N/mm**

M20 bolt single shear (4.6) ≈ 45 kN (bearing governs?) → $p = 45000/720 =$ **62 mm** → provide 60 mm c/c (≥ 2.5 d = 50 ✓, ≤ 200).

6. Plate Girders – Introduction

Large **I-section built up from plates** by welding (or bolting with angles) – used when span/loads exceed rolled sections: bridges, gantry girders, heavy floors, industrial buildings (spans 15–50 m ; depth ~ L/10 – L/12).

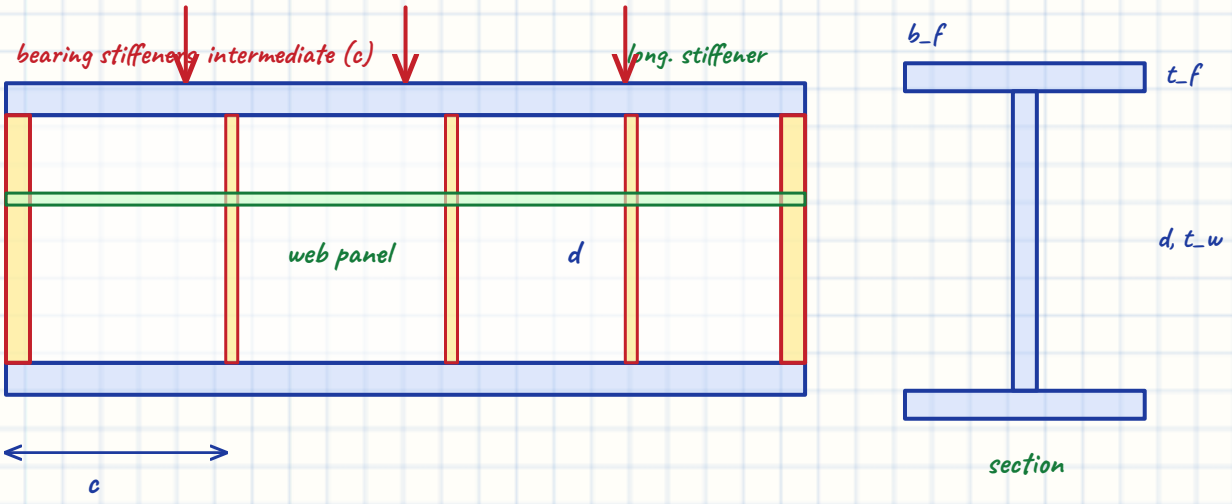


Plate girder – elevation with stiffeners and cross-section

Components: two **flanges** (carry bending: M/d), **web** (carries shear), **stiffeners** :

- **Bearing (load-carrying) stiffeners** – at supports & concentrated loads ; act as columns
- **Intermediate transverse stiffeners** – increase shear buckling resistance (spacing c)
- **Longitudinal stiffeners** – near compression flange (at 0.2 d) to reduce web slenderness
- **Flange splices, web splices**, end panels.

Advantages: material only where needed → lighter, economical, flexible depth & thickness. Disadvantages: fabrication & inspection, fatigue at stiffener welds.

7. Proportioning a Plate Girder

Depth of web d

Economical depth $d \approx (M_u / (f_y t_w))^{1/2} \cdot (...)$ – practical: **$d = L/10$ to $L/12$**. Check headroom & transport.

Web thickness t_w (IS 800 Cl. 8.6)

Serviceability (avoid flange buckling into web & excessive web distortions):

- No transverse stiffeners: **$d/t_w \leq 200\ \varepsilon$** ; with stiffeners the permissible d/t_w ↑ (up to ≈ 250-270 ε for $c < 1.5\ d$, and **$345\ \varepsilon^2$** limit to prevent flange buckling into the web). Choose $t_w \geq 8$ mm (min 6 mm – exposed).

Flange area A_f

Flanges carry the moment, web assumed to carry shear only (**flange area method**):

$$A_f \approx M_u \gamma_{m0} / (f_y\ d_f)\ ;\ d_f = \text{distance between flange centroids} \approx d + t_f$$

Adjust for web contribution: $A_f = M/(f_y\ d) - A_w/6$ (approx.).

- Flange outstand $b/t_f \leq$ **13.6 ε** (semi-compact) / 8.4ε (plastic) ; $b_f \approx d/5$ to $d/3$.
- Flange plate thickness, taper/curtail in 1–2 steps (flange plate changes) where M reduces.

8. Plate Girder – Strength Checks

(a) Bending (flange)

$$M_d = A_f\ d_f\ f_y / \gamma_{m0}\ (+ \text{web contribution for plastic design}) \geq M_u\ ;\ \text{for LTB use } \chi_{LT}\ (\alpha_{LT} = 0.49)$$

(b) Shear strength of web (IS 800 Cl. 8.4.2 – simple post-critical method)

$$V_d = V_{cr}/\gamma_{m0}\ ;\ V_{cr} = A_v\ \tau_b = d\ t_w\ \tau_b$$

Elastic critical shear stress:

$$\tau_{cr,e} = k_v\ \pi^2 E / [12\ (1 - \mu^2)\ (d/t_w)^2]$$

$k_v = 5.35$ (no intermediate stiffeners) ; $4 + 5.35/(c/d)^2$ for $c/d < 1$; $5.35 + 4/(c/d)^2$ for $c/d \geq 1$

$$\lambda_w = \sqrt{(f_{yw} / (\sqrt{3}\ \tau_{cr,e}))}$$

λ_w	τ_b
≤ 0.8	$f_{yw}/\sqrt{3}$
$0.8 - 1.2$	$[1 - 0.8(\lambda_w - 0.8)]\, f_{yw}/\sqrt{3}$
≥ 1.2	$f_{yw}/(\sqrt{3}\,\lambda_w^2)$ $(= \tau_{cr,e})$

Tension-field method (Cl. 8.4.2.2) gives higher V_d (needs end panel + stiffeners anchoring).

(c) Interaction M–V

If $V > 0.6\, V_d$: M reduced (as beams) ; for web carrying part of moment.

(d) Stiffeners

- **Intermediate stiffener**: min $I_s \ge 0.75\, d\, t_w^3$ $(c/d \ge \sqrt{2})$; $\ge 1.5\, d^3 t_w^3/c^2$ $(c/d < \sqrt{2})$; outstand $\le 12\, t_s\, \varepsilon$; spacing $c \le 3d$ (1.5 d typical).
- **Bearing stiffener** : act as a **column** of cross-section = stiffener plates + 20 t_w web strip each side ; effective length 0.7 d ; $F_x \le P_d$; also bearing check on the stiffener contact area (ground ends) : $F_{psd} = A_p\, f_y/\gamma_{m0} \ge R_u$. Weld to web for full reaction.
- **Web–flange weld** : $q = V\, A_f\, \bar{y}\, /\, I$ → weld size (two fillet lines).

EXAMPLE 4 – SIMPLY SUPPORTED PLATE GIRDER, L = 20 M

w_u = 50 kN/m (including self-weight) → $M_u = 50 \times 20^2/8 =$ **2500 kN·m** ; $V_u =$ **500 kN**.

Depth $d = L/11 \approx$ **1800 mm** ; $t_w = 10$ mm → $d/t_w = 180 < 200\, \varepsilon$ $(=200)$ ✓

Flange area $A_f = 2500\times10^6 \times 1.1 / (250 \times 1810) \approx$ **6080 mm²** → provide **400 × 16** ($A = 6400$ mm²) ; outstand $= (400 - 10)/2 = 195 \to 195/16 = 12.2 < 13.6$ ✓

Shear : stiffener spacing $c = 1800$ $(c/d = 1)$ → $k_v = 5.35 + 4/1 =$ **9.35**

$\tau_{cr,e} = 9.35 \times \pi^2 \times 2\times10^5 / [12 \times 0.91 \times 180^2] =$ **52.2 MPa**

$\lambda_w = \sqrt{(250/(\sqrt{3} \times 52.2))} =$ **1.66 ≥ 1.2** → $\tau_b = \tau_{cr,e} = 52.2$

$V_{cr} = 1800 \times 10 \times 52.2 = 939$ kN → $V_d = 939/1.1 =$ **854 kN > 500** ✓

Web–flange weld: $q = V\, A_f\, \bar{y}/I$ → $I \approx 2\times6400\times905^2 + 10\times1800^3/12 = 1.05\times10^{10} + 4.9\times10^9 = 1.5\times10^{10}$ mm⁴ ; $\bar{y} \approx 905$ → $q = 500\times10^3\times6400\times905/1.5\times10^{10} = 193$ N/mm → two fillet welds: 96.5 N/mm each → **s = 3–4 mm → min size 5 mm.**

Bearing stiffener at support: $R_u = 500$ kN → 2 plates 110 × 12 + web strip 2×(20×10) …

Order for exam answer: loads → depth → web thickness → flange → check bending → shear (k_v, λ_w, τ_b) → stiffeners (intermediate, bearing) → welds → splices & sketch.

9. More on Laterally Unsupported Beams

When can a beam be treated as laterally supported? (IS 800 Cl. 8.2.2)

- Compression flange **fully restrained** against lateral deflection & rotation (e.g. cast in concrete slab / connected to slab with shear connectors, sheeting with positive connection)
- Hollow (box, tube) sections and I-sections bending about the **minor axis**
- Closely spaced **bracings** so that $\lambda_{LT} \le 0.4$ (i.e. L_{LT}/r_y small)

- Slab bearing on top flange **with friction only is NOT enough** – need positive connection.

Effective length L_{LT} for LTB (Table 15 – selected)

Restraint at supports	L_{LT}
Compression flange fully restrained laterally; ends torsionally restrained (simply supported beam)	1.0 L
Both flanges free to rotate on plan (fork support), no intermediate restraints	1.2 L
Partial torsional restraint at ends	1.2 L – 1.0 L
Cantilever – fixed end restrained, free end unrestrained	1.0 L (load at tip: 1.4–2.5 L)
Load destabilising (acts at top flange, free to move)	add 20 % (×1.2)

L = span or distance between lateral restraints.

Procedure for design of unsupported beam (trial & error)

- 1. Find M_u. Assume $f_{bd} \approx 100$–130 MPa → $Z_p(req) = M_u/f_{bd}$ → select section.
- 2. Compute L_{LT}/r_y, h_f/t_f → $f_{cr,b}$ → λ_{LT} → φ_{LT} → χ_{LT} → f_{bd}.
- 3. $M_d = \beta_b Z_p f_{bd} \geq M_u$. If not, choose a deeper/wider flange section or reduce L_{LT} by bracing.
- 4. Check shear, deflection, web crippling exactly as laterally supported beam.

> Quick check: ISMB 300 (L=4 m) → $M_d \approx 84$ kN·m ; if bracing at mid-span reduces L_{LT} to 2 m → f_{bd} rises (λ_{LT} falls to ≈ 0.75) ⇒ $M_d \approx 120$ kN·m. **Bracing is cheaper than a bigger section!**

EXAMPLE 5 – HIGH SHEAR CHECK

Simply supported beam ISMB 400, span 3 m, central factored point load 400 kN (laterally supported). Properties: h = 400, t_w = 8.9 mm, $Z_p \approx 1176$ cm³ , Z_e = 1022.9 cm³, b_f = 140, t_f = 16.0.

$M_u = 400 \times 3/4 =$ **300 kN·m** ; $V_u =$ **200 kN**.

$V_d = 400 \times 8.9 \times 250/(\sqrt{3} \times 1.1) =$ **467 kN** ; $0.6\ V_d = 280 > 200$ → low shear.

$M_d = 1176 \times 10^3 \times 250/1.1 =$ **267 kN·m** < 300 × ⇒ section fails in bending. Try ISMB 450 ($Z_p \approx 1533$ cm³ → M_d = 348 kN·m) ✓.

If V were 300 kN (> 0.6 V_d) → $\beta = (2 \times 300/467 - 1)^2 = 0.081$ → $M_{dv} = M_d - 0.081 (M_d - M_{fd})$ – small reduction.

10. Bearing Stiffeners & End Bearing of Beams

Where a heavy reaction / concentrated load acts and web crippling / buckling check fails → provide **load-carrying stiffeners** (plates welded each side of the web, fitted tightly to the flange).

Design of a load-bearing stiffener

- 1. Required **bearing area** : $A_{req} = R_u \gamma_{m0} / f_y$ (contact area of stiffener ends with the loaded flange, excluding corner snips).
- 2. Choose stiffener plates (2 No.) : outstand $b_s \le$ **12 $t_s \varepsilon$** ; thickness $t_s \ge b_s/12\varepsilon$... (usually ≥ 8–10 mm)
- 3. **Column check**: effective section = 2 stiffeners + web strip of width **20 t_w** on each side (or 25 t_w for end stiffener) ; effective length = **0.7 d** (both ends fixed) → r, λ, f_{cd} → $P_d = f_{cd} A_{eff} \ge R_u$.
- 4. **Weld** stiffener to web : length d, force R_u → fillet weld length 2(d − 2×15) ... size from $q = R_u/(\Sigma L_w)$.

EXAMPLE 6 – BEARING STIFFENER AT SUPPORT OF A PLATE GIRDER

R_u = 500 kN ; web 1800 × 10, flange 400 × 16 ; f_y = 250. Take 2 plates 120 × 12, ground to bear.

(i) Bearing area (excluding snip 15 mm) : 2 × (120 − 15) × 12 = 2520 mm² → capacity = 2520 × 250/1.1 = **573 kN** > 500 ✓

(ii) Column : A_{eff} = 2 × 12 × 120 + (2 × 20 × 10 + 10) × 10 = 2880 + 4100 = **6980 mm²**

$I_{web\ axis} \approx 12 \times (2 \times 120 + 10)^3/12 = 12 \times 250^3/12 = 1.56\times10^7$ mm⁴ → $r = \sqrt{(1.56\times10^7/6980)} = 47.3$ mm

$KL/r = 0.7 \times 1800/47.3 =$ **26.6** → $f_{cd} \approx$ **195 MPa** (class c) ⇒ $P_d \approx 195 \times 6980 =$ **1360 kN** > 500 ✓

(iii) Weld to web : force 500 kN over 2 × 1700 mm length → 147 N/mm → minimum 5 mm fillet ✓.

Web splices & flange splices

- **Flange splice** : cover plates with area ≥ flange area; bolts/welds for flange force.
- **Web splice** : plates both sides ; carries shear + moment share $M_w = M \times (I_w/I)$.
- Locate splices where moments are low ; full penetration butt welds are best (shop).

11. Plate Girder – Design Steps (Complete Flow)

- 1. **Data** : span L, loads (UDL, concentrated), f_y ; reactions R_u, M_u, V_u (include self-weight ≈ 2-3 kN/m initial).
- 2. **Web depth** d = L/10 to L/12 (≥ economical depth) ; **web thickness** t_w from $d/t_w \le 200\varepsilon$ (no stiffener) or from shear; minimum 8 mm.
- 3. **Flanges** : $A_f = M_u \gamma_{m0}/(f_y\, d)$; choose b_f & t_f ($b_f \approx 0.3$-$0.4\ d$; $t_f \ge b_f/(2\times13.6\varepsilon)$) ; extra **cover plate** where M is high (curtail).
- 4. **Moment of inertia & Z_e** ; classify ; $M_d \ge M_u$ (check LTB if L_{LT} large).
- 5. **Web stiffener spacing** c : choose c ≈ d ... 1.5 d (end panel c ≤ d) ; shear V_d from simple post-critical method (or tension field) ≥ V_u at each panel.
- 6. **Shear-moment interaction** in panels with high V and M.

- 7. **Bearing stiffeners** at supports & point loads ; **intermediate stiffeners** $I \geq$ limit ; connect with welds.
- 8. **Web–flange connection** weld : $q = V A_f \bar{y} / I$.
- 9. **Splices**, deflection check ($\delta = 5wL^4/384EI$ with service loads $\leq L/300$ or $L/360$).
- 10. Draw elevation & cross-sections: label d, t_w, b_f, t_f, c, stiffener sizes, weld sizes.

Tension field action (idea only)

After the web buckles in shear (**post-buckling**), it can still carry additional shear as **diagonal tension** between stiffeners; flanges & stiffeners act like a **Pratt truss** (diagonal tension band + stiffeners strut). Gives ≈ 20–50 % more shear. Needs strong end panels & anchorage.

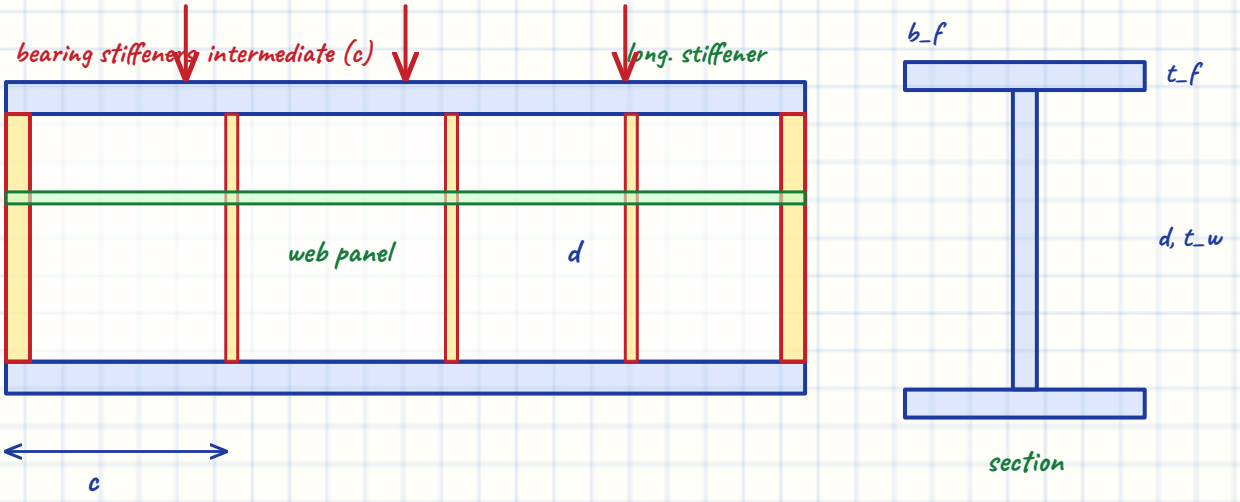


Plate girder – shear resisted by web, moment by flanges

12. Rolled Beam vs Built-up vs Plate Girder

Feature	Rolled beam	Built-up (cover plates)	Plate girder
Span range	up to ~ 10-12 m	~12-20 m	15 – 50+ m
Depth	≤ 600 mm (ISMB 600)	600-900	1–3 m
Fabrication	none	simple	complex
Design basis	table M_d	M_d + connection of plate	flange + web + stiffener design
Typical use	floors, purlins	heavy floors	bridges, gantry, heavy industrial

Important formulae – Unit III (quick sheet)

$$M_d = \beta_b Z_p f_y/\gamma_{m0} \le 1.2 Z_e f_y/\gamma_{m0}$$

$$V_d = h t_w f_y/(\sqrt{3}\gamma_{m0})\ ;\ \text{low shear if } V \le 0.6 V_d$$

$$\delta = 5wL^4/(384 E I)$$

$$f_{bd} = \chi_{LT} f_y/\gamma_{m0}\ ;\ M_d = \beta_b Z_p f_{bd}$$

$$\tau_{cr,e} = k_v \pi^2 E/[12(1-\mu^2)(d/t_w)^2]\ ;\ V_{cr} = d t_w \tau_b$$

$$q = V A \bar{y} / I \text{ (shear flow)}\ ;\ F_w = (b_1 + n_2) t_w f_y/\gamma_{m0}$$

Important exam questions

- 1. Differentiate laterally supported and unsupported beams. Explain LTB and factors.
- 2. Design a simply supported beam (L = 6 m, w = 25 kN/m DL+LL) with full lateral support.
- 3. Design a beam with L_{LT} = 3 m – check χ_{LT}.
- 4. Design a welded plate girder for span 18 m : proportioning, stiffeners, weld.
- 5. Explain web buckling, web crippling and design of bearing stiffeners.
- 6. Design cover plates for I-beam and their bolted connection ; curtailment.